Question	Answer	Marks
1	**Defining the problem**	
	vary r and measure v **or** r is the independent variable and v is the dependent variable	**1**
	keep x constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • one end of spring resting against block clamped to bench using G-clamp • light gate positioned at P • light gate connected to timer • apparatus shown on bench • labels for light gate **and** P **and** at least one other label from bench, block, stand, spring, ball, timer	**1**
	method to determine r, e.g. use calipers or micrometer to measure diameter d **and** $r = d / 2$	**1**
	description of method to determine v, use diameter of ball (to interrupt beam) ÷ measured time at light gate positioned at P	**1**
	instrument to determine x, e.g. rule(r) or calipers	**1**
	Method of Analysis	
	plot a graph of (2 lg v) or (lg v^2) against lg r **or** plot a graph of (lg v) against (lg r) or equivalent, e.g. (ln v) against (ln r)	**1**
	$n = -$gradient for (2 lg v) or (lg v^2) against lg r **or** $n = -2 \times$ gradient for (lg v) against lg r	**1**

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Question	Answer	Marks
1	$Y = \frac{\rho \times 10^{y\text{-intercept}}}{kx^2}$ for (2 lg v) or (lg v^2) against lg r **or** $Y = \frac{\rho \times 10^{2 \times y\text{-intercept}}}{kx^2}$ for (lg v) against lg r	**1**
	Additional detail including safety considerations	**6**
	D1 precaution to <u>prevent ball leaving bench</u>, e.g. screens around apparatus / cushions on bench (to stop the ball)	
	D2 keep k **and** ρ constant	
	D3 description of method to determine k, e.g. add mass to spring and $k = mg$ / extension **or** use newton meter to measure force applied to spring and k = force / extension **or** take several readings of force and extension, plot a force–extension graph and k = gradient	
	D4 description of experimental method to determine ρ, e.g. measure mass of ball using a balance **and** $\rho = \frac{m}{\frac{4}{3}\pi r^3}$	
	D5 repeat measurements of <u>diameter</u> or <u>d</u> in different directions <u>and</u> determine the average value of d	
	D6 method to keep x constant, e.g. use a pin / ruler / card to indicate the starting point each time <u>to keep x constant</u>	
	D7 x = original length of spring – <u>compressed</u> length of spring	
	D8 adjust (vertical) position of light gate so that the diameter of (each) ball cuts the beam	
	D9 repeat experiment for the same value of r **and** determine the average v	

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Question	Answer	Marks
1	D10 relationship valid <u>if</u> a straight line is produced (with y-intercept = $\lg\left(\frac{Ykx^2}{\rho}\right)$ or $\frac{1}{2}\lg\left(\frac{Ykx^2}{\rho}\right)$). Do not accept line through the origin.	

Question	Answer	Marks
2(a)	gradient = $H\lambda_0$ y-intercept = λ_0	**1**
2(b)	$\frac{d}{c}$ / 10^{15} s (1.6 or 1.60) ± 0.40 (3.47 or 3.467) ± 0.40 (4.83 or 4.833) ± 0.40 (6.00 or 6.000) ± 0.40 (9.50 or 9.500) ± 0.40 (12.5 or 12.50) ± 0.40 Values of $\frac{d}{c}$ correct as shown above.	**1**
	Uncertainties in $\frac{d}{c}$ correct as shown above.	**1**

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